1. Identity

Document	STAR-PKG-STAR-GASDYNAMICS-EN · v2.1 · 2026-11-01
Package	star-gasdynamics
Version	0.1.0
Capability	star.propulsion.gasdynamics
Maturity	RELEASED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_gasdynamics

2. Licence and obligations

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3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import
hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())"
star_gasdynamics-0.1.0-py3-none-any.whl
```

FILE	SHA-256
star_gasdynamics-0.1.0-py3-none-any.whl	a3c3838c7e1bdce3ac656e04ee39e2ded6ed9db015ddddc2f1a6176dcfb37215
star_gasdynamics.py	6a7dab40d85811ce82ac4e3f45eef5f09477444c7e35eca8a43878abad1cd09c
star_gasdynamics-0.1.0-source.zip	ffd066f926e7056173253e6c77209f43bdf89e94ceff7546bc5e6979c87c7e90

5. What it does, and its limits

Isentropic flow ratios, Mach number from area ratio on either branch and normal-shock jump of a calorically perfect gas; within 1.2e-15 of scikit-aero and fluids on 600 cases for the relations they provide; reproduces the NACA Report 1135 tables; density, temperature and total-pressure ratios across the shock rest on the published table and the conservation laws only.

A calorically perfect gas only: no real-gas effects, no dissociation, no variable gamma, so the results are not valid for hot hypersonic flows whatever the Mach limit accepted. rho2/rho1, T2/T1 and p02/p01 have NO library lineage: no installed library computes them; they rest on the published table and on the conservation laws. scikit-aero's inverse of the area ratio did not converge, or converged to the other branch, in about a quarter of the 600 cases (451 and 460 usable roots per branch): those cases were not compared; recorded as a candidate observation about a third-party library, not investigated and not reported. No oblique shocks, no Prandtl–Meyer expansion, no Fanno or Rayleigh flow.

- `isentropic(mach, gamma)`: T/T0, p/p0, rho/rho0 and A/A* of a calorically perfect gas (NACA Report 1135). Reproduces the published table for gamma = 1.4 at Mach 0.5, 2, 3 and 5.
- `mach_from_area_ratio(area_ratio, gamma, supersonic)`: the inverse on the branch asked for, by bisection to the last bit. `normal_shock(mach1, gamma)`: M2 and the four ratios; reproduces the published table at Mach 2, 3 and 5 and satisfies the conservation of mass, momentum and energy across the shock.
- Measured on 600 cases (Mach 0.01 to 20, gamma 1.05 to 1.67): T/T0, p/p0, A/A*, M2 and p2/p1 within 1.2e-15 relative of scikit-aero; the pressure-temperature relation and the sonic ratios

within 1.2e-15 of fluids; the Mach number from the area ratio within 2e-12 of scikit-aero where its root finder converged to the right branch.

- Every function returns finite floats (A/A* is infinite at Mach 0) or raises `ValueError`: non-numeric, boolean, NaN or infinite input, Mach outside [0, 50] (shock: [1, 50]), gamma outside [1.01, 3], an area ratio below 1 or beyond the branch, `supersonic` not a bool.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_gasdynamics_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-gasdynamics}: Isentropic flow and normal shock of a perfect gas},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_gasdynamics/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

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